

# PAPER\_855: Pseudo-Monopole 26-State Vacuum Density Progression

**Author:** Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok\_share\_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025) **Calculator:** PseudoMonopole26StateVacuumDensityCalc (CP4 #439) **CVW:** v2.0.0 compliant

---

## Abstract

We derive the full 26-state pseudo-monopole vacuum density progression within UQFF. The angular spacing  $\delta_n = (2\pi)^{n/6}$  defines the pseudo-monopole geometry at each quantum state  $n = 1 \dots 26$ , while the vacuum density ratio  $\rho_{vac,[UA]:SCm} = 1e-23 (0.1)^n * \exp(-[SSq]n/26) \exp(-(\pi - t))$  governs the energy landscape. At  $n=1, t=0$ :  $\delta_1 \sim 1.047$  rad,  $\rho_{vac} \sim 9.63e-25$  J/m<sup>3</sup>. The exponential suppression across 26 states spans over 25 orders of magnitude in vacuum density.

---

## 1. Core Equations

- $\delta_n = (2\pi)^{n/6}$  -- pseudo-monopole angular spacing
  - $\rho_{vac,[UA]:[SCm]}(n, t) = 1e-23 * (0.1)^n * \exp(-[SSq]*n/26) * \exp(-(\pi - t))$
  - $n = 1$ :  $\delta_1 \sim 1.047$  rad,  $\rho \sim 9.63e-25$  J/m<sup>3</sup>
  - $n = 26$ :  $\delta_{26} \sim (2\pi)^{(26/6)}$ ,  $\rho \sim 1e-23 * 1e-26 * \exp(-SSq) \sim$  vanishing
- 

## 2. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies\_\*.csv data flow.

---

## 3. Source Data

- **File:** grok\_share\_589f6949-6fe9.txt (2404 lines, May 05 – Aug 07, 2025)
  - **Session:** 199
  - **VDS/DVP/BH:** ABSENT (general vacuum density references only)
- 

## References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. – Electroweak neutron production (LENR)
3. Kepler Mission DR25 – 4,034 candidates, 2,335 confirmed planets
4. Hubble Heritage Team / A. Nota (ESA/STScI) – Westerlund 2 / NGC 346 imaging
5. UQFF Calibration:  $\kappa=0.0005/\text{day}$ ,  $[SSq]=0.57$ ,  $\beta_i \sim 0.603$